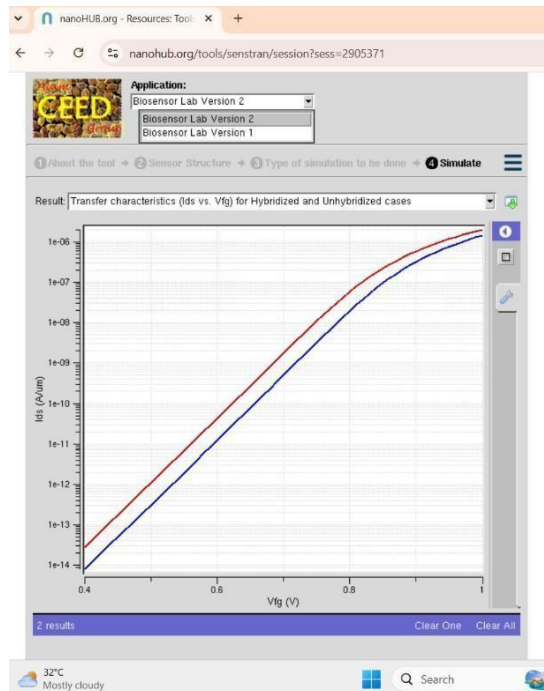


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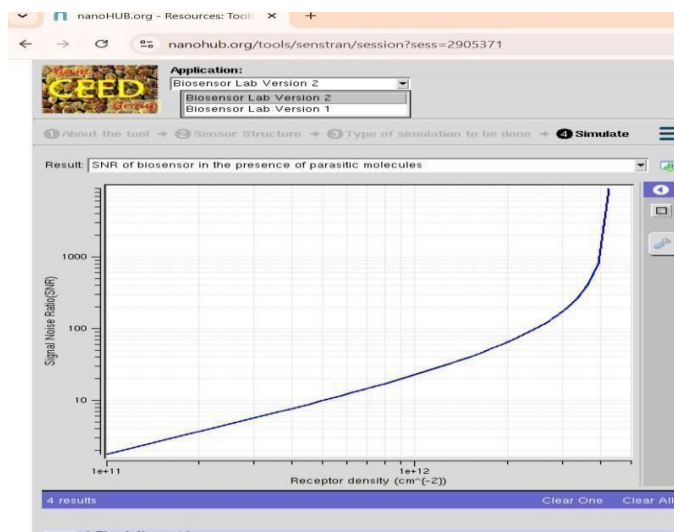
Exp. 9. Biosensor and Bio-FET characteristics using NanoHub

Case Study 1: Comparative Analysis of Hybridized and Unhybridized BioFET Transfer Characteristics for Enhanced Biosensing Performance



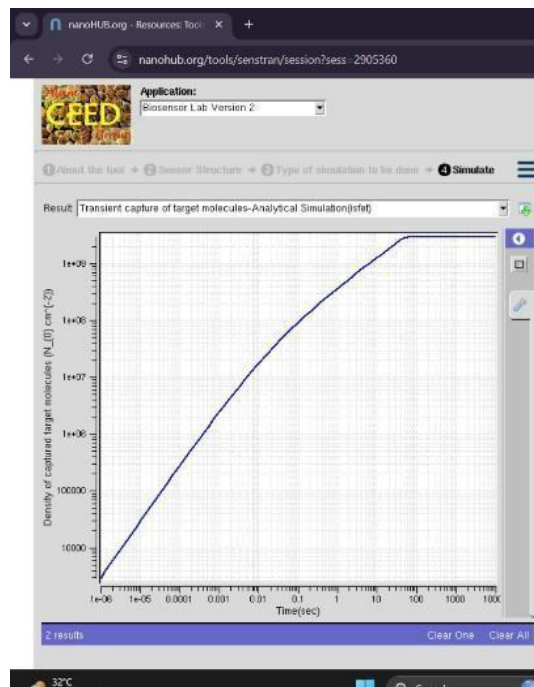
Outcome: On an industrial scale the film deposition is done by magnetron sputter deposition; on a laboratory scale molecular beam epitaxy, pulsed laser deposition and electron beam physical vapor deposition are also utilized. The junctions are prepared by photolithography.

Case Study 2: Evaluation of Signal-to-Noise Ratio (SNR) Variation with Receptor Density in the Presence of Parasitic Molecules



Outcome: The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ($2000 \text{ cm}^2/\text{Vs}$) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

Case Study 3: Transient Analysis of Target Molecule Capture Dynamics in a BioFET-Based Biosensor



Outcome: The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ($2000 \text{ cm}^2/\text{Vs}$) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

